

Do corporate climate ratings advance or hinder the Net-Zero transition?

Four essays in sustainability strategy and economics

PhD Candidate in Engineering economics

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of Denmark

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Introduction



CentraleSupélec

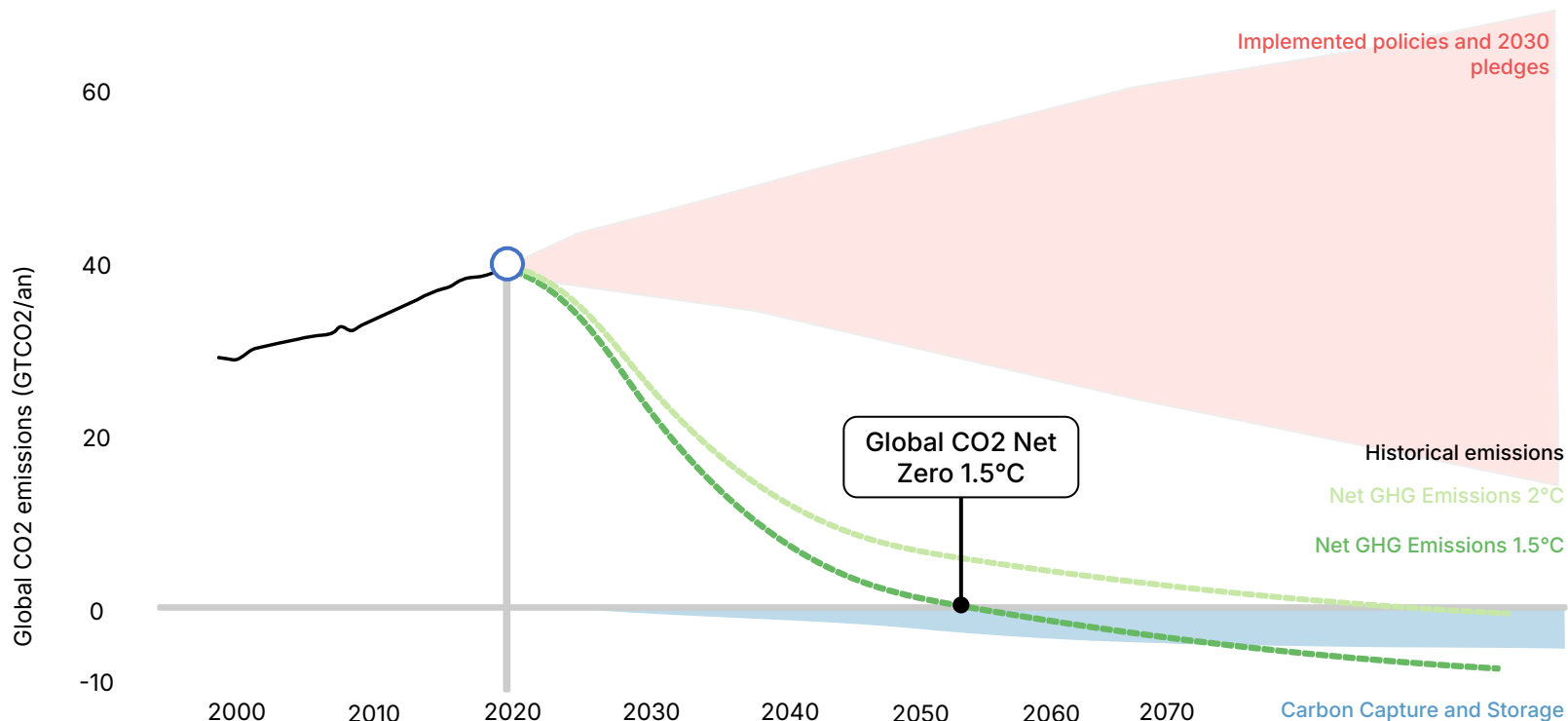
université
PARIS-SACLAY



greenly

State of global climate action

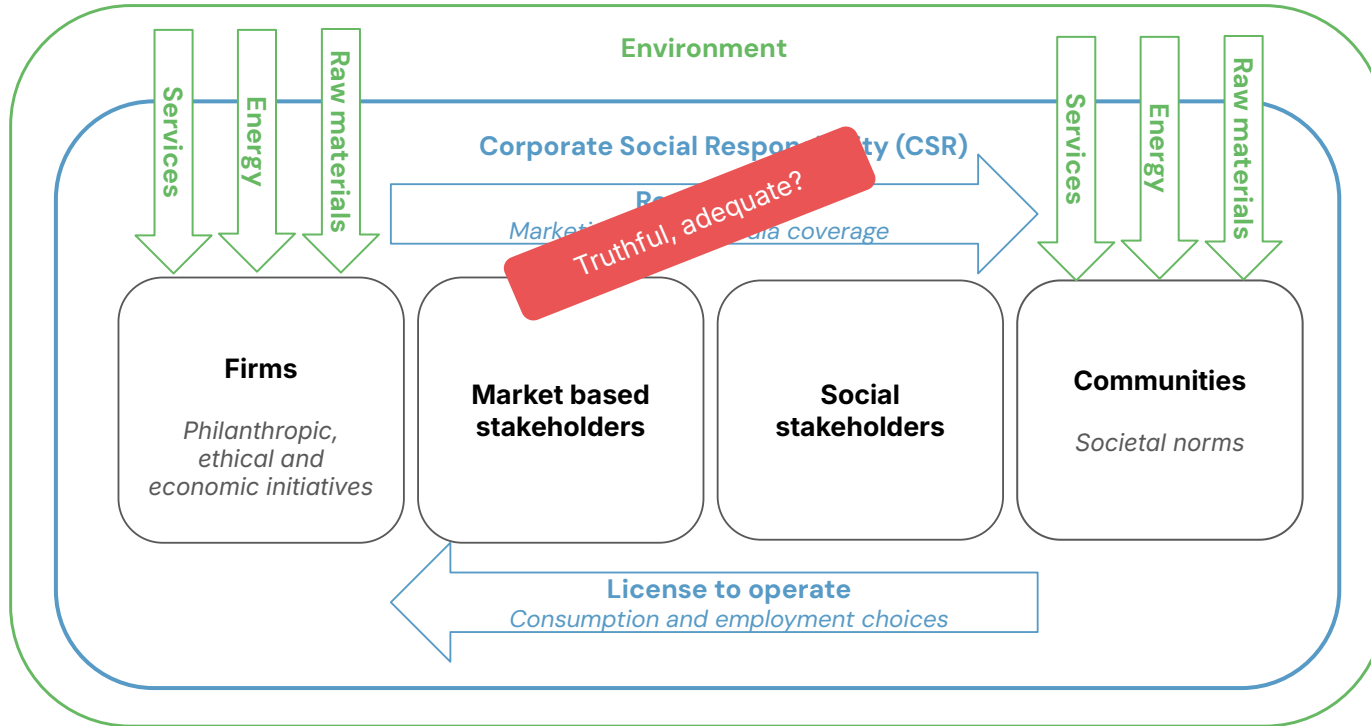
Projected emissions pathways are insufficient to respect the Paris Agreement



Adapted from IPCC, 2022: Summary for Policymakers [P.R. Shukla, J. Skea, A. Reisinger [...]]. In: Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change.

Beyond regulation, voluntary climate action

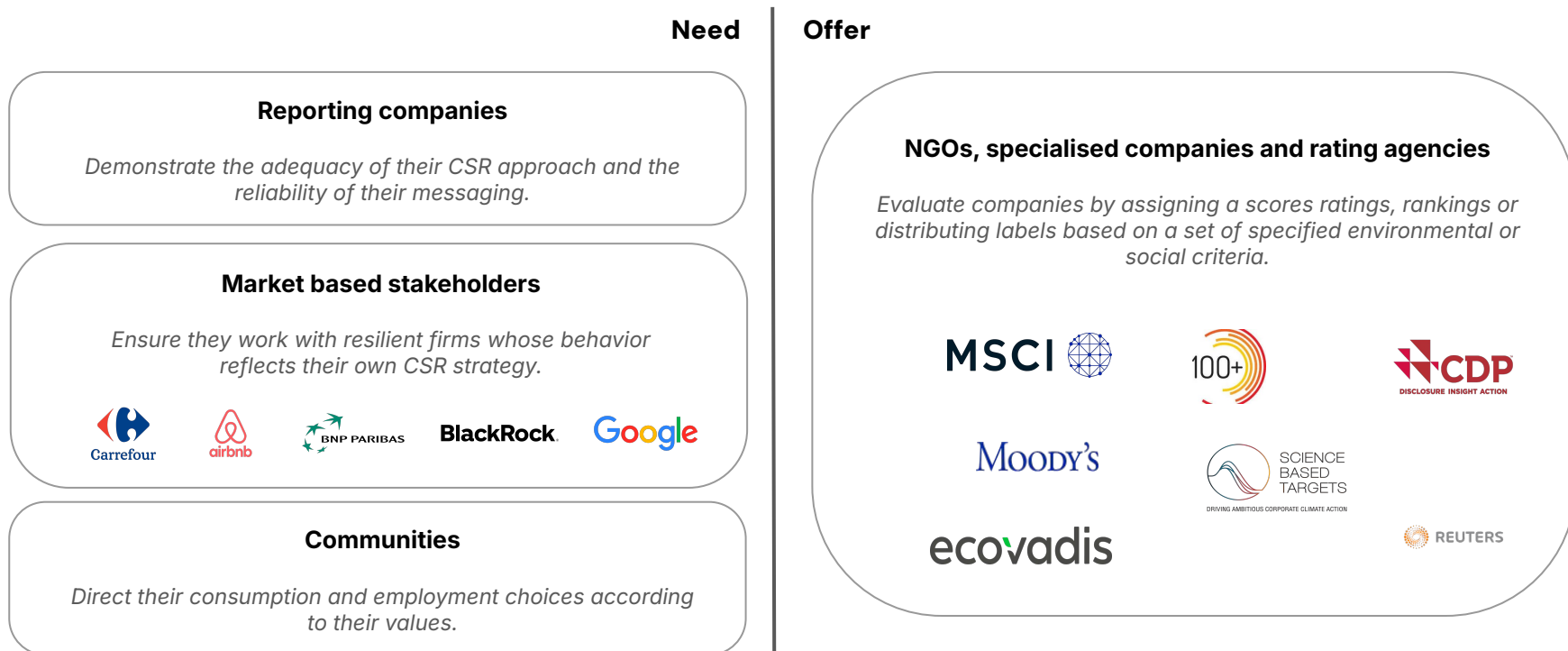
Firms are expected to adapt their behaviour to societal expectations and must adapt to a changing environment



Selected elements from Carroll, A. B. (2021), *Business & Society*, 60(6), 1258–1278; Dowling, J., & Pfeffer, J. (1975), *The Pacific Sociological Review*, 18(1), 122–136; & Hart, Stuart L., *The Academy of Management Review*, vol. 20, no. 4, 1995, pp. 986–1014.

Scoring systems as CSR firm evaluation facilitators

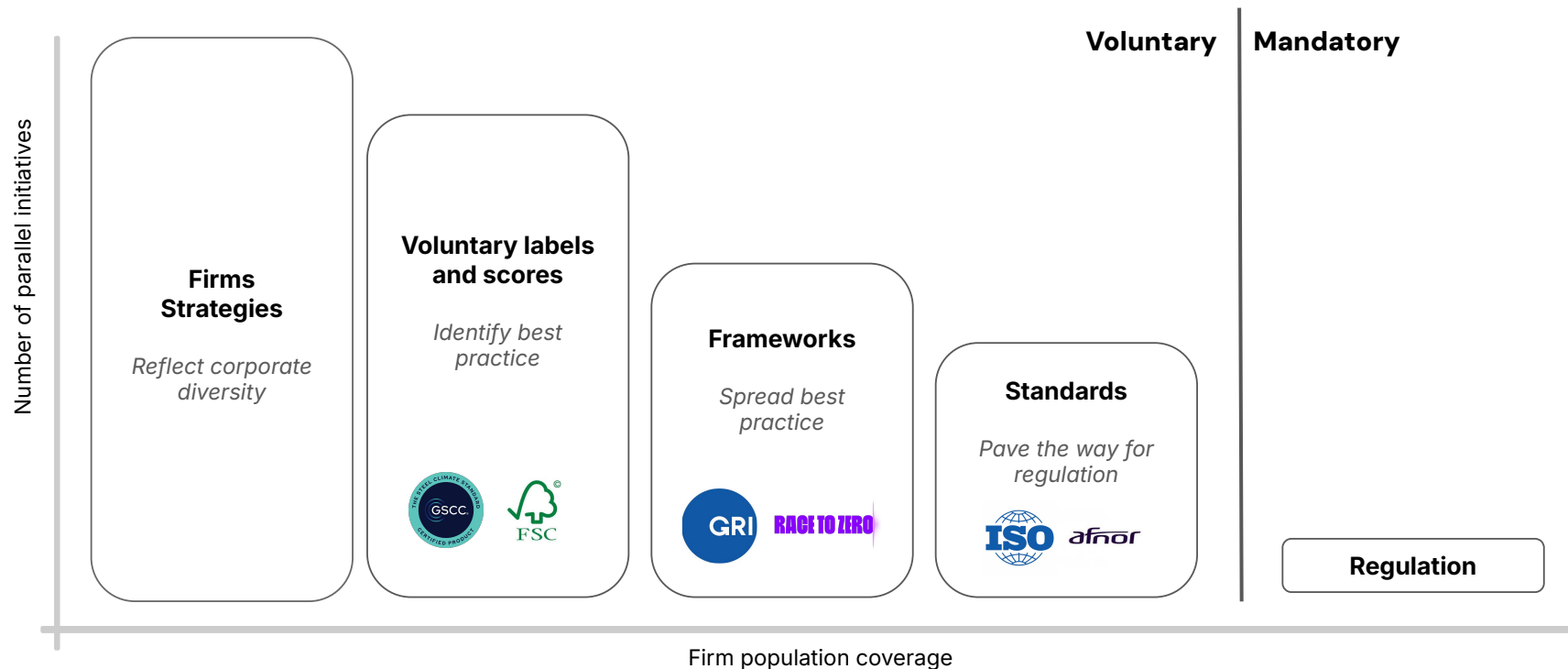
Environmental, social and governance (ESG) evaluations proliferated since the 1990 to facilitate sustainable investment decision-making



Selected elements from Kohler, M., Da Costa, P. et al., Renewable and Sustainable Energy Reviews, Volume 226, Part C, 2026, 116352, ISSN 1364-0321; Seow, R. Y. C. (2025), Journal of Environmental Management, 394, 127468; Avetisyan, E., & Ferrary, M. (2013), Journal of Business Ethics, 115(1), 115–133.

Self-regulation driving policy

Framework developers must strike a balance between ambition and scalability to ensure the success of their initiatives



Adapted from Hale, T. (2022, December). The net-zero governance conveyor belt. Kleinman Center for Energy Policy. Evidence for stakeholder expectation management can be found on the CDP, GHG Protocol and SBTi websites.

How much hope to place on such initiatives?

A mixed literature outlook

Are corporate climate performance measures appropriately framing sustainability?

Simple vs double materiality

Are climate performance measures driving sustainability improvements?

*Mixed empirical results on emissions mitigation
Multiplicity of actors undermining effectiveness
Potential to delay effective regulation*

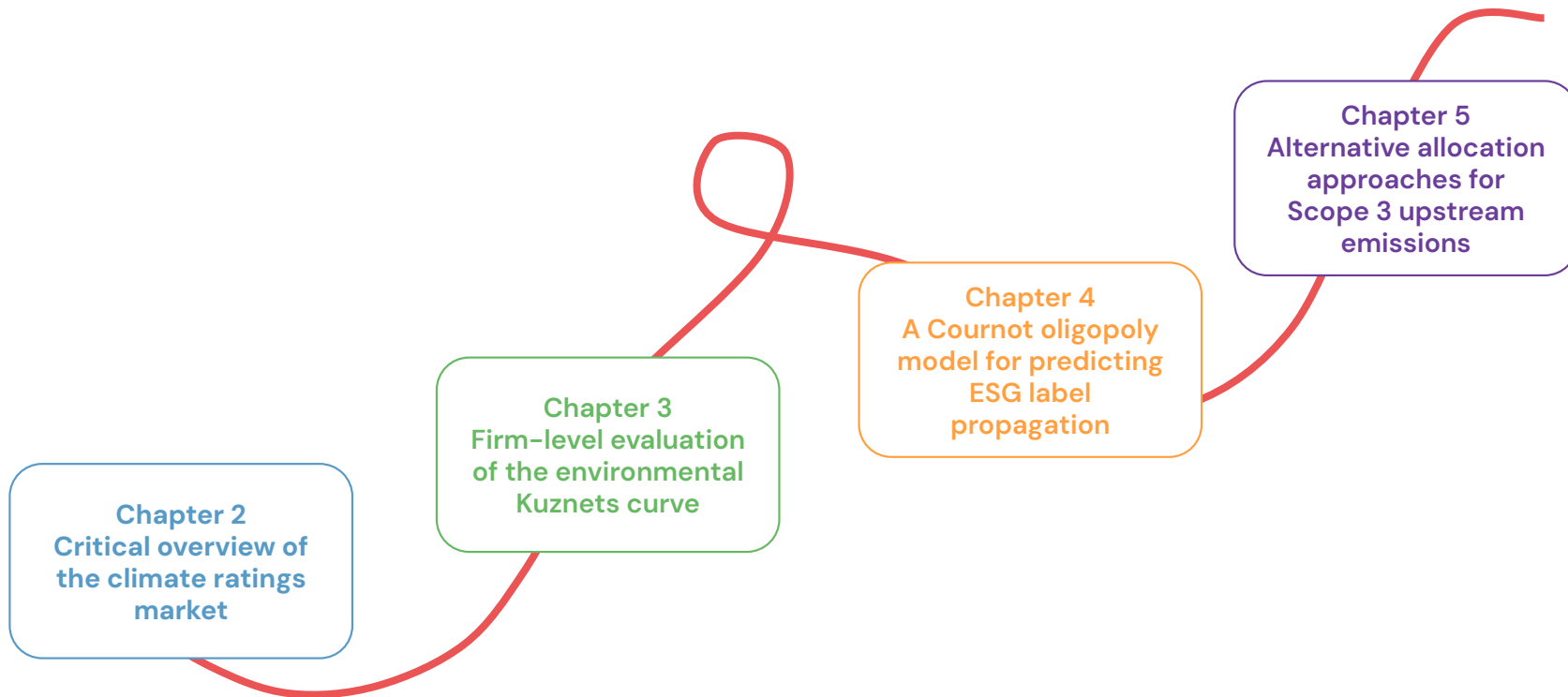
How are corporate climate performance measures shaping futures societies and economies?

Arbitrary and opaque arbitrage between stakeholders with potential power to define futures

Tilsted, J., Palm, E., et al. (2023). Energy Research & Social Science, 103, 103229.; Fischer, C., & Lyon, T. (2019). Journal of the Association of Environmental and Resource Economists, 6(3), 461–501.; Brécard, D. (2014). Resource and Energy Economics, 37, 64–84.; Klaaßen, L., Lohmüller, C., et al. (2024). (L.Hsueh, Ed.). PLOS Climate, 3(11), e0000458.; Bjørn, A., Lloyd, S., et al. (2022). Nature Climate Change, 12(6), 539–546.; Malhotra, N., Monin, B., et al. The American Political Science Review 113, no. 1 (2019): 19–37.

Presentation overview

Four diversified parts with a common thread



Ariane textiles

A common thread in this presentation

Midsized,
historical,
regional textile
assembly
company

Neither an
industry leader
nor a laggard

Market based stakeholder

*Supplier engagement efforts
to encourage proactive
climate strategies.*

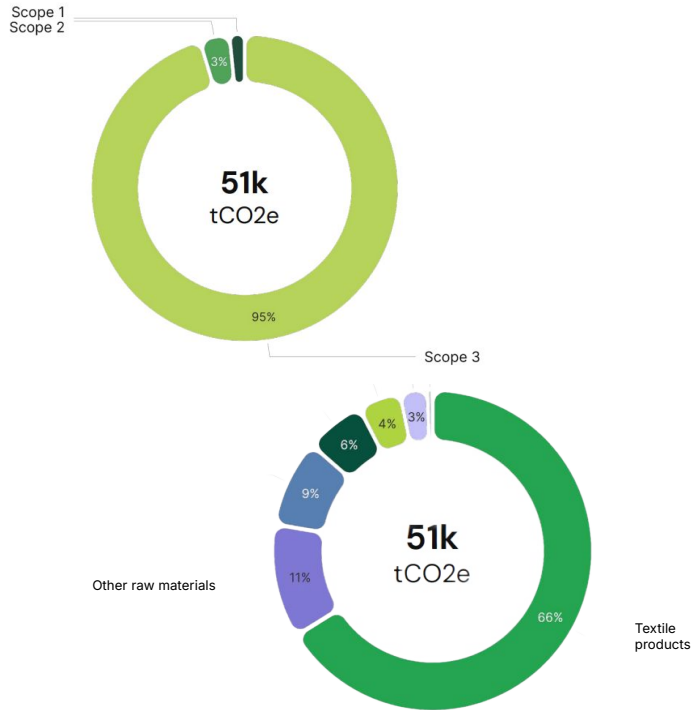


Communities

*Increasing internal questioning
on the impact of the firm.*



GHG Protocol compliant 2024 GHG Assessment



Chapter 2

A review of corporate climate ratings:
Assessing divergence from scientific
expectations

*Marine Kohler, Pascal Da Costa, François
Cluzel, Loïc Umbricht,
Renewable and Sustainable Energy
Reviews, Volume 226, Part C, 2026,
116352, ISSN 1364-0321.*



CentraleSupélec



Research questions

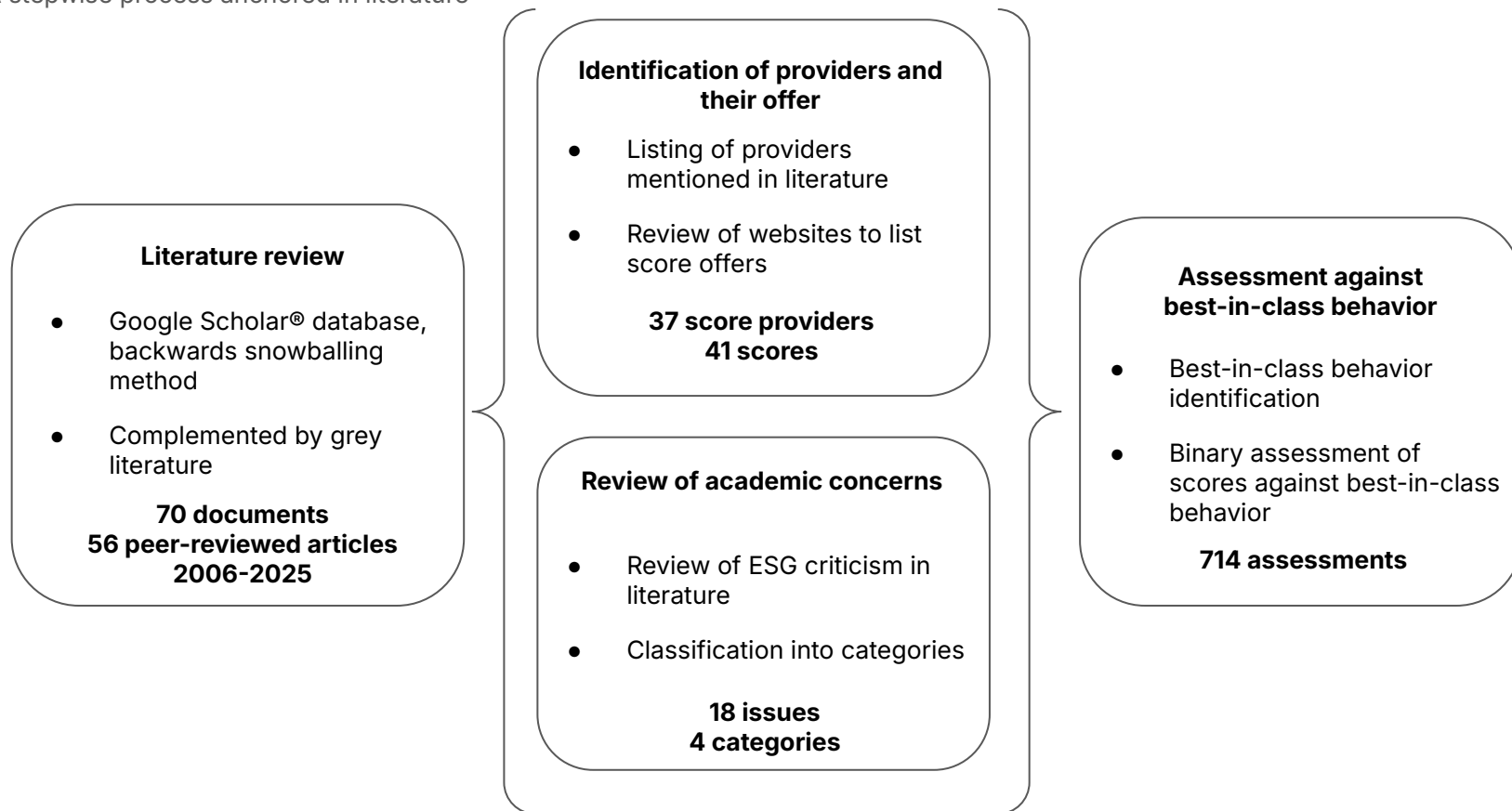
An initial assessment of 2024 market offers



- What are the **current corporate climate rating systems on the market?**
- What defines a **trustworthy** corporate sustainability measurement system?
- **How trustworthy are current market offers?**

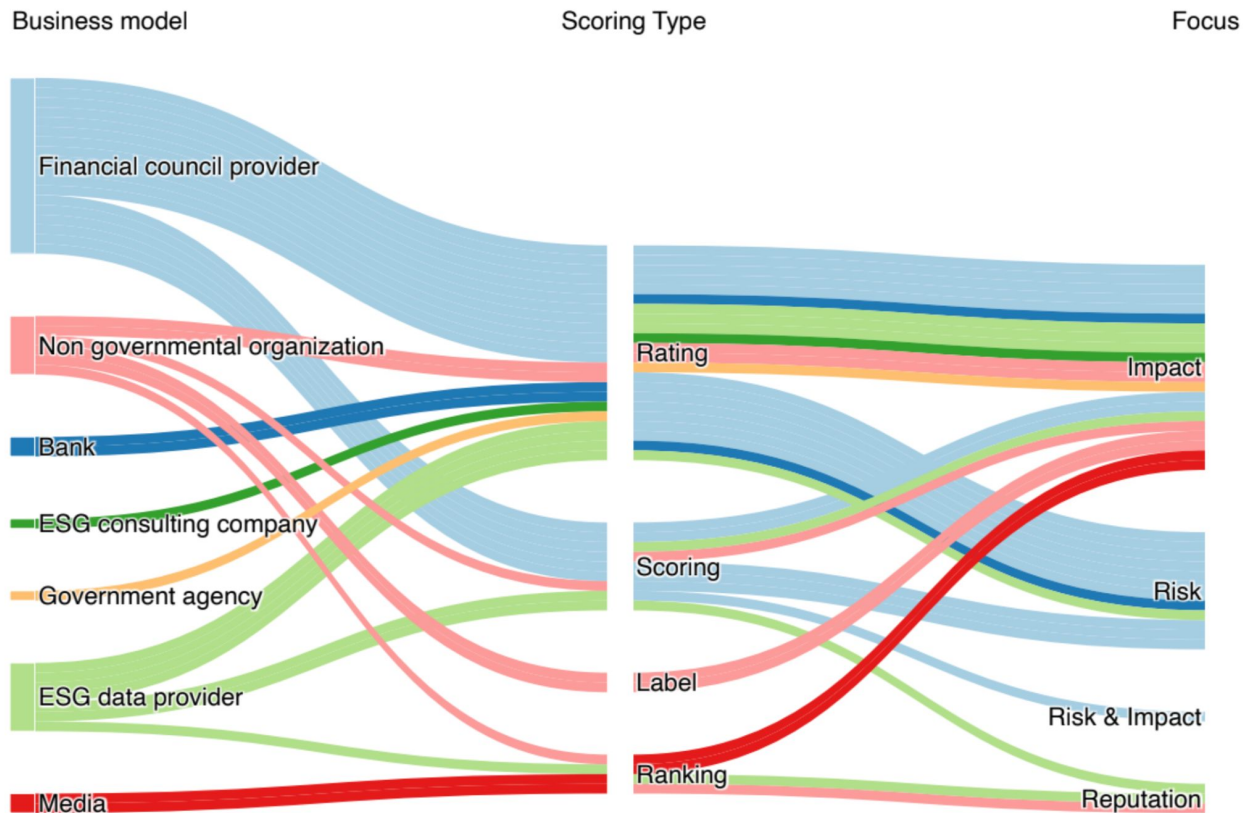
Methodology

A stepwise process anchored in literature



Climate performance measurement systems

Repartition of scoring types and focuses



Academic concerns on ESG scoring systems

An overview



Categories

Criticisms

Reliability - Capacity to deliver a consistent and comparable assessment

Accuracy - Capacity to provide correct information

Fairness - Capacity to show impartial and just treatment

Effectiveness - Capacity to drive sustainability improvements

Selected references : Windolph, S. E. (2011). Journal of Environmental Sustainability, 1(1); Escrig-Olmedo, E., Fernández-Izquierdo, M. Á., Ferrero-Ferrero, I., Rivera-Lirio, J. M., & Muñoz-Torres, M. J. (2019). Sustainability, 11(3), 915; Chatterji, A., Durand, R., et al. (2016). Strategic Management Journal, 37(8), 1597–1614. OECD. (2022, June), OECD Business and Finance Policy Papers No. 06.

SBTi, a pillar of corporate climate performance measurement

And thus, Ariane textile's next logical step

Corporate climate assessments

MSCI



Venture
Climate Alliance

Moody's

REUTERS

ecovadis

CDP
DISCLOSURE INSIGHT ACTION

WE MEAN
BUSINESS
COALITION



NET ZERO
TRACKER



SCIENCE
BASED
TARGETS

DRIVING AMBITIOUS CORPORATE CLIMATE ACTION

Chapter 3

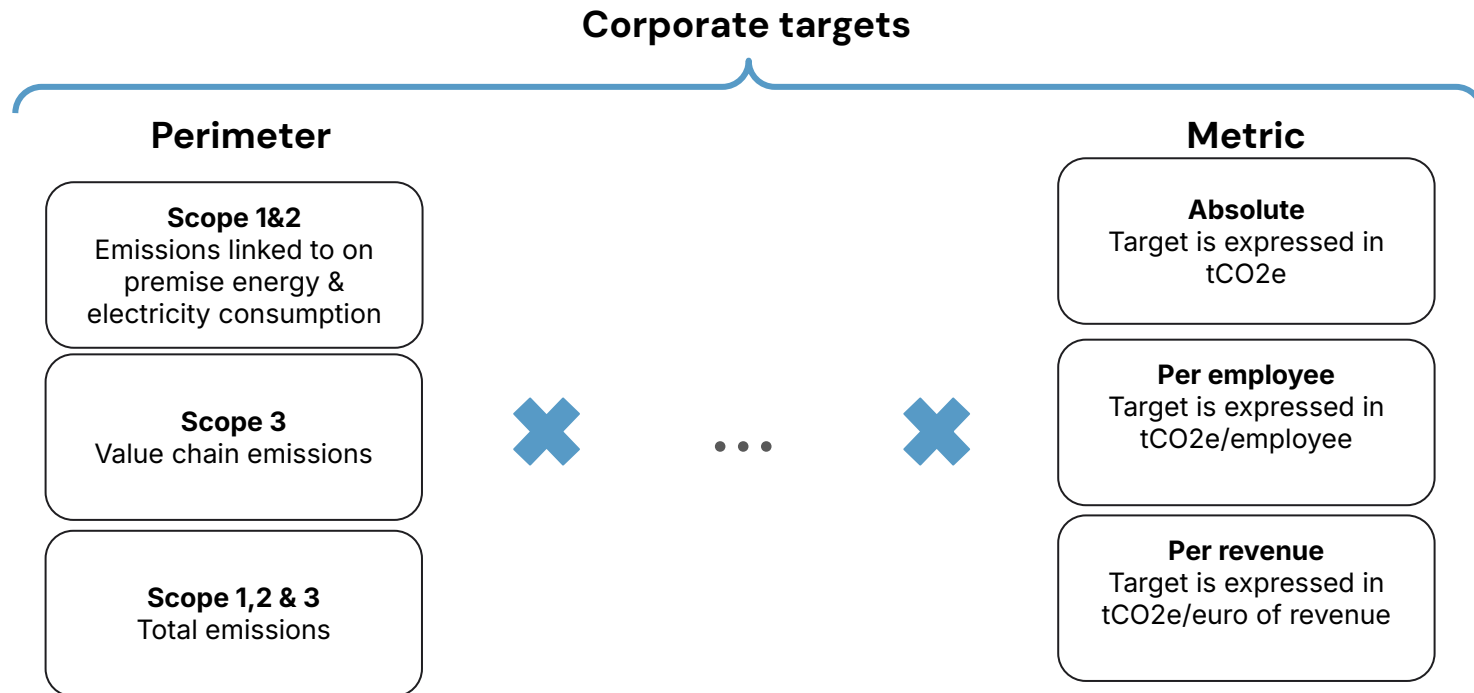
Firm-level evaluation of the environmental Kuznets curve:
Thresholds in cross-sectional data from small, medium and mid-sized enterprises

Marine Kohler, Natalia Costa i Coromina, Pascal Da Costa, Peter Fox Penner, François Cluzel, Under review in the Journal of Cleaner Production.



Options for SBTi compliant targets

What should Ariane textile choose?



Options for SBTi compliant targets

What should Ariane textile choose?

Corporate targets

Perimeter

Scope 1&2

Emissions linked to on
premise energy &
electricity consumption

Scope 3

Value chain emissions

Scope 1, 2 & 3

total emissions

Metric

Per employee

Target is expressed in
tCO₂e/employee

Per revenue

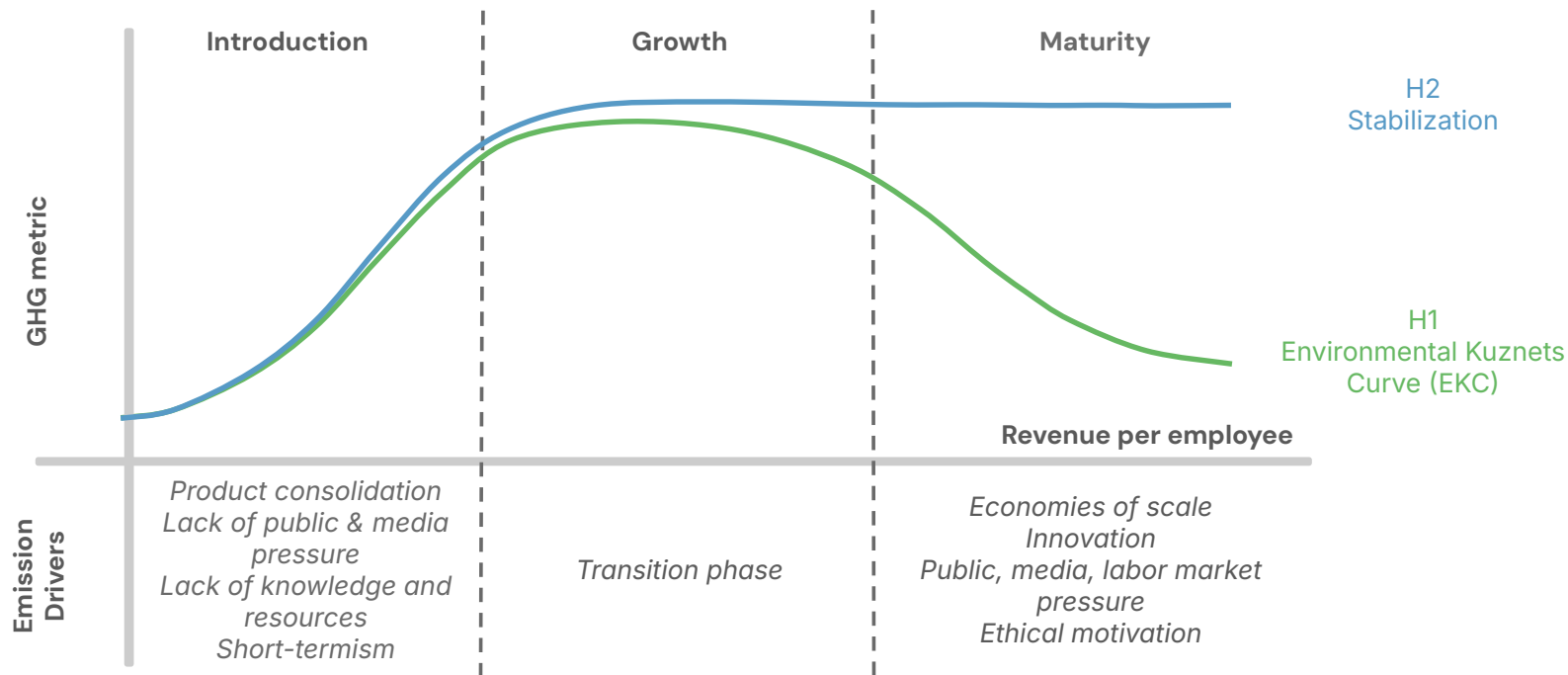
Target is expressed in
tCO₂e/euro of revenue

Limited data available to guide companies across industries towards fair, sufficient yet achievable targets



Novel hypotheses

Drivers and barriers to climate action are unevenly distributed across firm maturity levels



Drivers and barriers are adapted from Furlan Alves, M. B., Lopes De Sousa Jabbour, A. B. et al. (2019), Journal of Organizational Change Management, 32(7), 687–708. and Johnson, M. P., Rötzel, T. S. et al. (2023), Management Review Quarterly, 73(2), 921–954.

Model specifications

Competing polynomial and threshold models

Database

greenly

inpi
INSTITUT NATIONAL
DE LA PROPRIÉTÉ
INDUSTRIELLE



GREENHOUSE
GAS PROTOCOL

SMEs and
mid-sized
companies

Cross
sectional
dataset

H1. Polynomial model

$$Y = \beta_0 + \beta_1 \cdot (R/E)^2 + \beta_2 \cdot (R/E) + \sum_{i=1}^n \alpha_i \cdot X_i + \epsilon$$

- **Y** is an **emission metric** over a certain perimeter of emissions
 - R/E denotes **revenue per employee**
- **X** denote the company size, industry, country, climate strategy and year **control features**, preselected using a **Lasso** regression
- Fitted using **OLS** with **firm and NACE division clustered** standard errors.

H2. Endogenous threshold model model

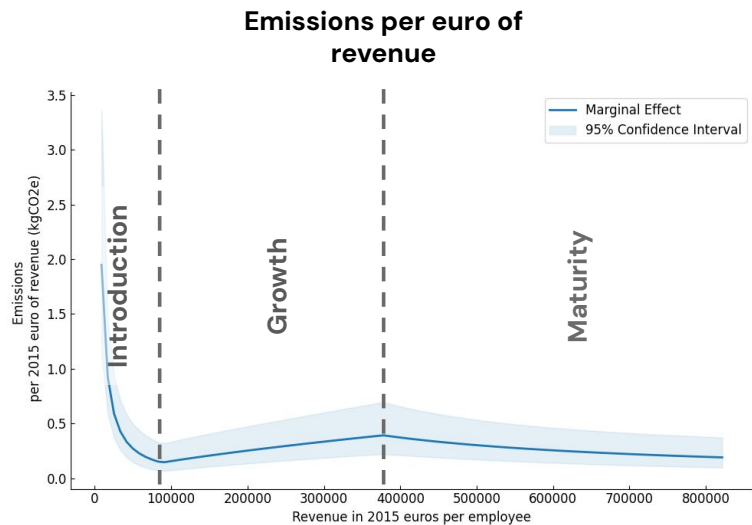
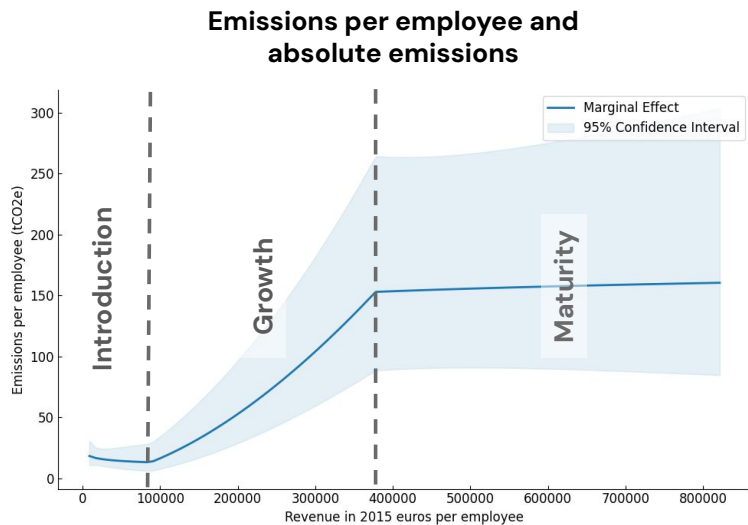
$$Y = \beta_0 + \beta_1 \cdot (R/E) + \beta_2 \cdot (R/E)_T + \sum_{i=1}^n \alpha_i \cdot X_i + \epsilon$$

- **T** is an **endogenous threshold**
- $(R/E)_T$ captures any **additional effect when revenue per employee exceeds the threshold**.

Methodology sources include Muggeo, V. M. R. (2003). Estimating regression models with unknown break-points. *Statistics in Medicine*, 22(19), 3055–3071, <https://doi.org/10.1002/sim.1545> and Belloni, A., & Chernozhukov, V. (2013). Least squares after model selection in high-dimensional sparse models. *Bernoulli*, 19(2). <https://doi.org/10.3150/11-BEJ410>.

Results

Threshold patterns across emissions metrics with significant influence on emission levels



Across emissions
scopes &
subscopes

Robust to post
stratification in
France

Robust to omitted
variable bias

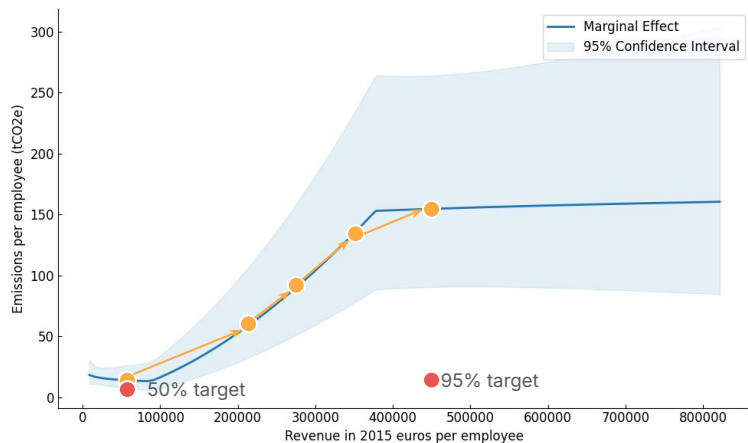
Limited reverse
causality concerns

Robust to the
introduction of ESG
maturity control
variables

Practical consequences

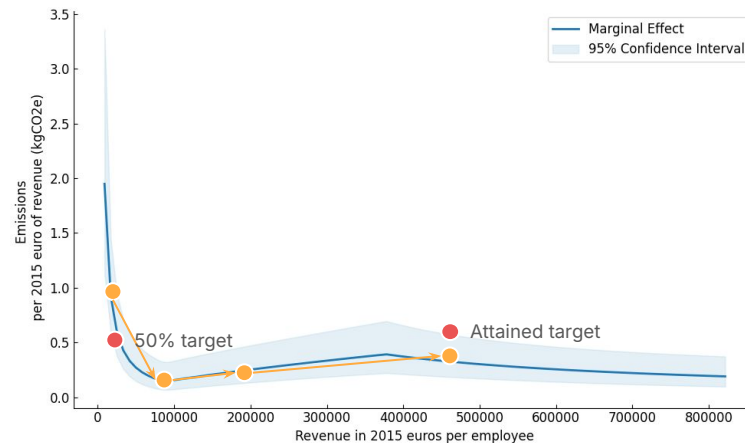
Unattainable and manipulable targets?

Emissions per employee and absolute emissions



Unattainable target

Emissions per euro of revenue



Manipulable target

Chapter 4

A Cournot oligopoly model for
predicting ESG label propagation under
cost heterogeneity

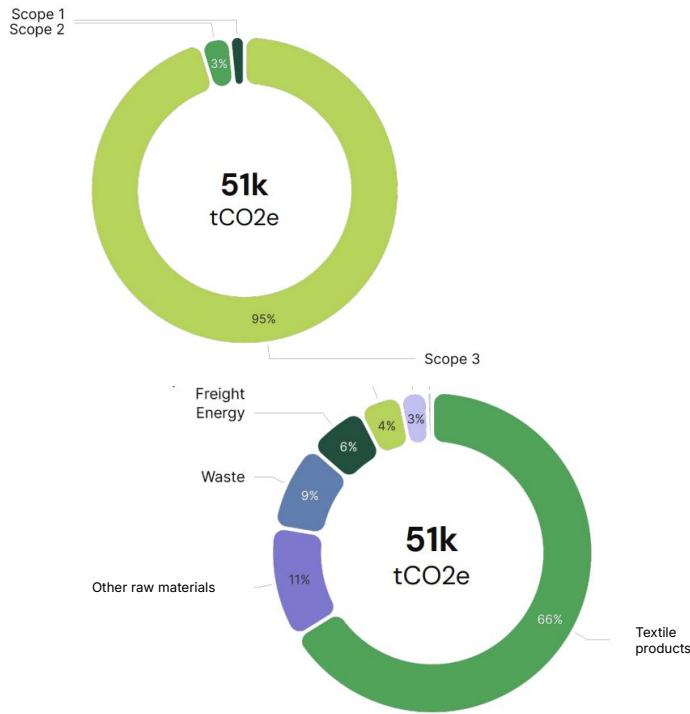
*Marine Kohler, Pascal Da Costa, Paul de
Kerret*



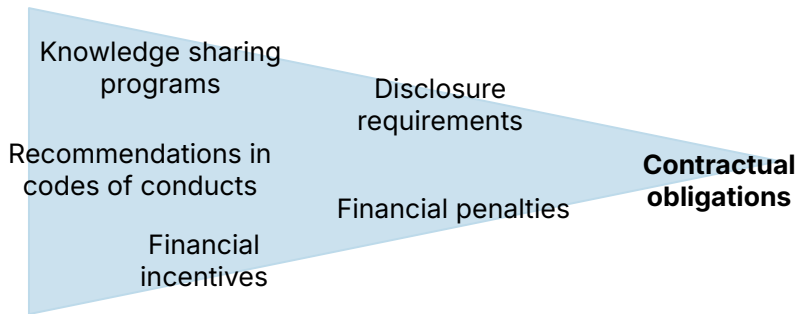
Supplier engagement at the heart of climate strategies

Is the hope placed on contractual obligations legitimate?

GHG Protocol compliant 2024 GHG Assessment



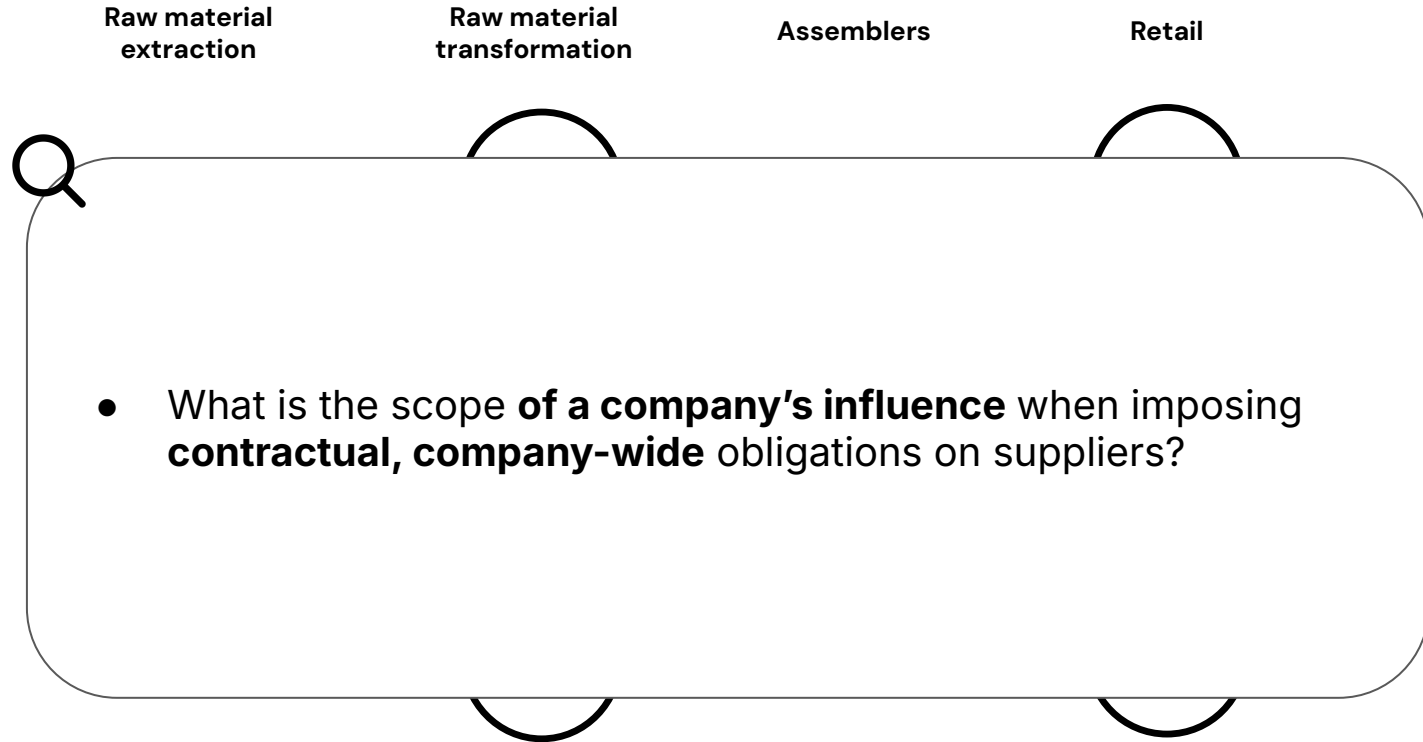
Industry recommendations for supplier engagement



CDP & HSBC. (2024, October). Strengthening the chain - transform the norm. CDP. ; We mean business coalition. (2025, April). Building resilient supply chains: Getting the most out of supplier engagement. We mean business coalition. SBTi. (2025, July). Engaging supply chains in the decarbonization journey. SBTi.

Research question

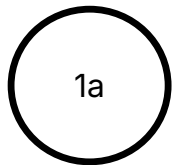
Schematic model



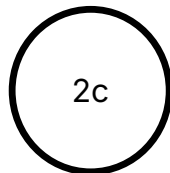
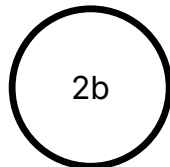
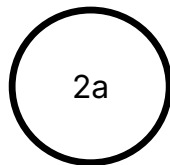
Research question

Schematic model

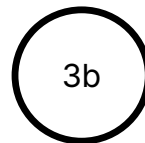
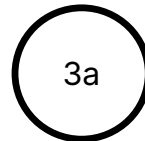
**Raw material
extraction**



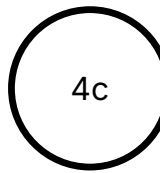
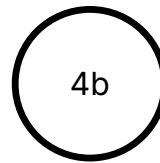
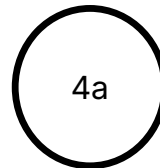
**Raw material
transformation**



Assemblers

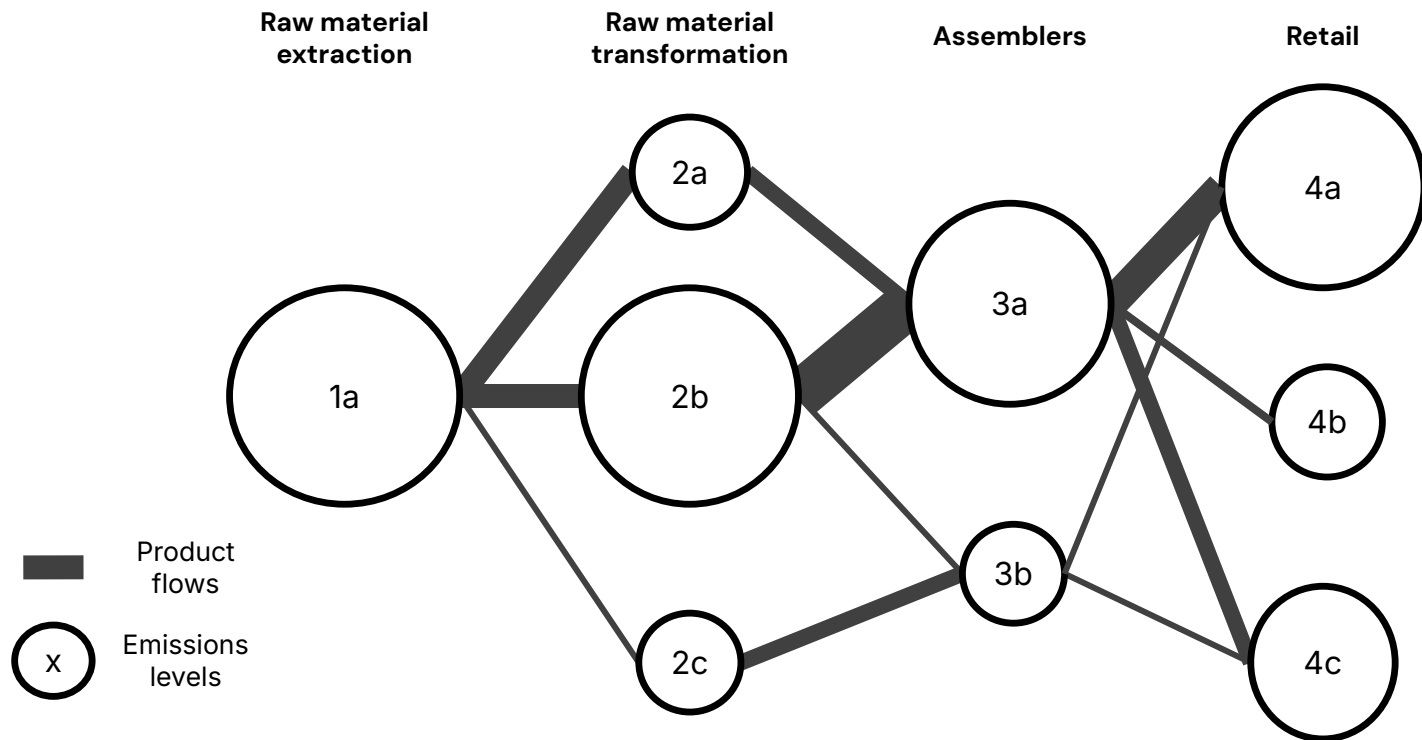


Retail



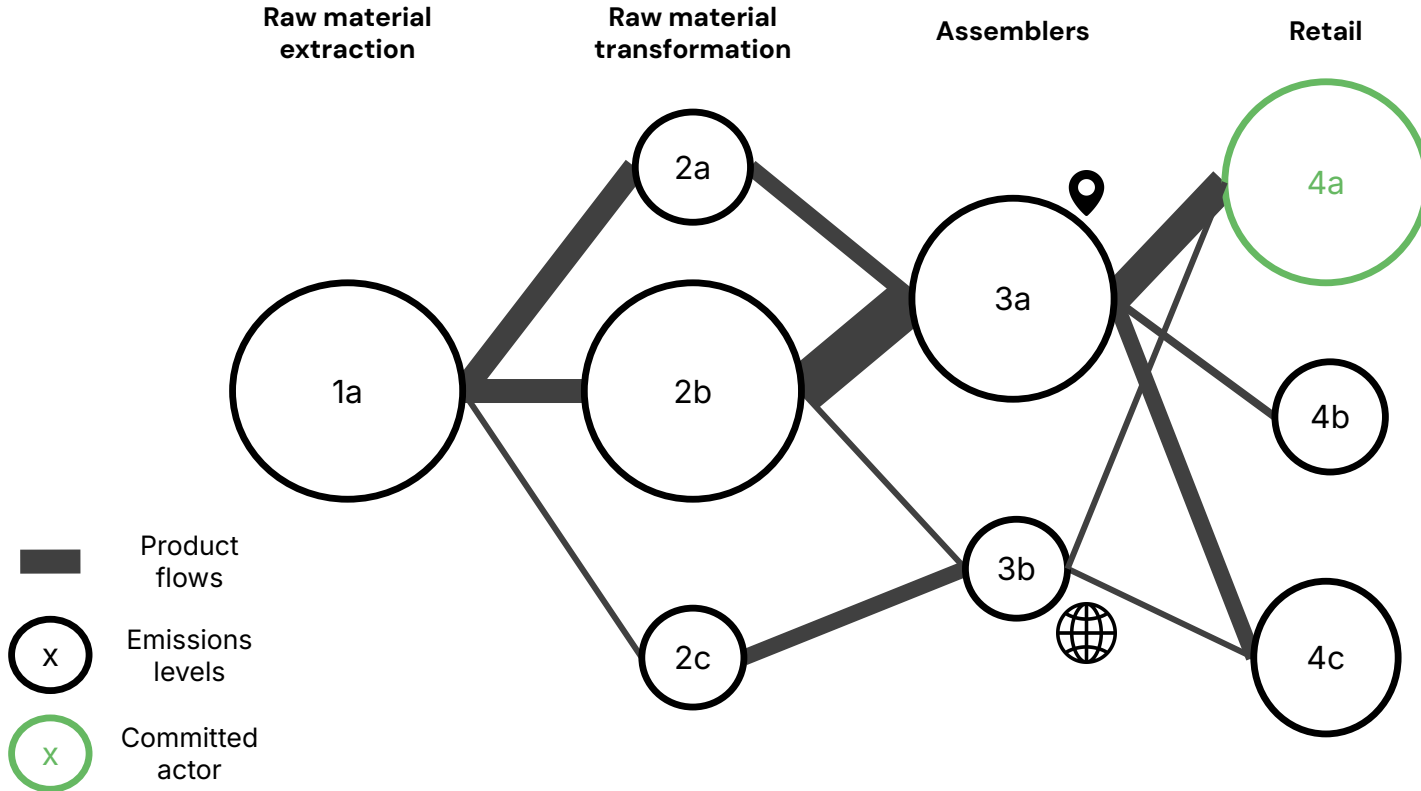
Model overview

Initial state under Cournot competition



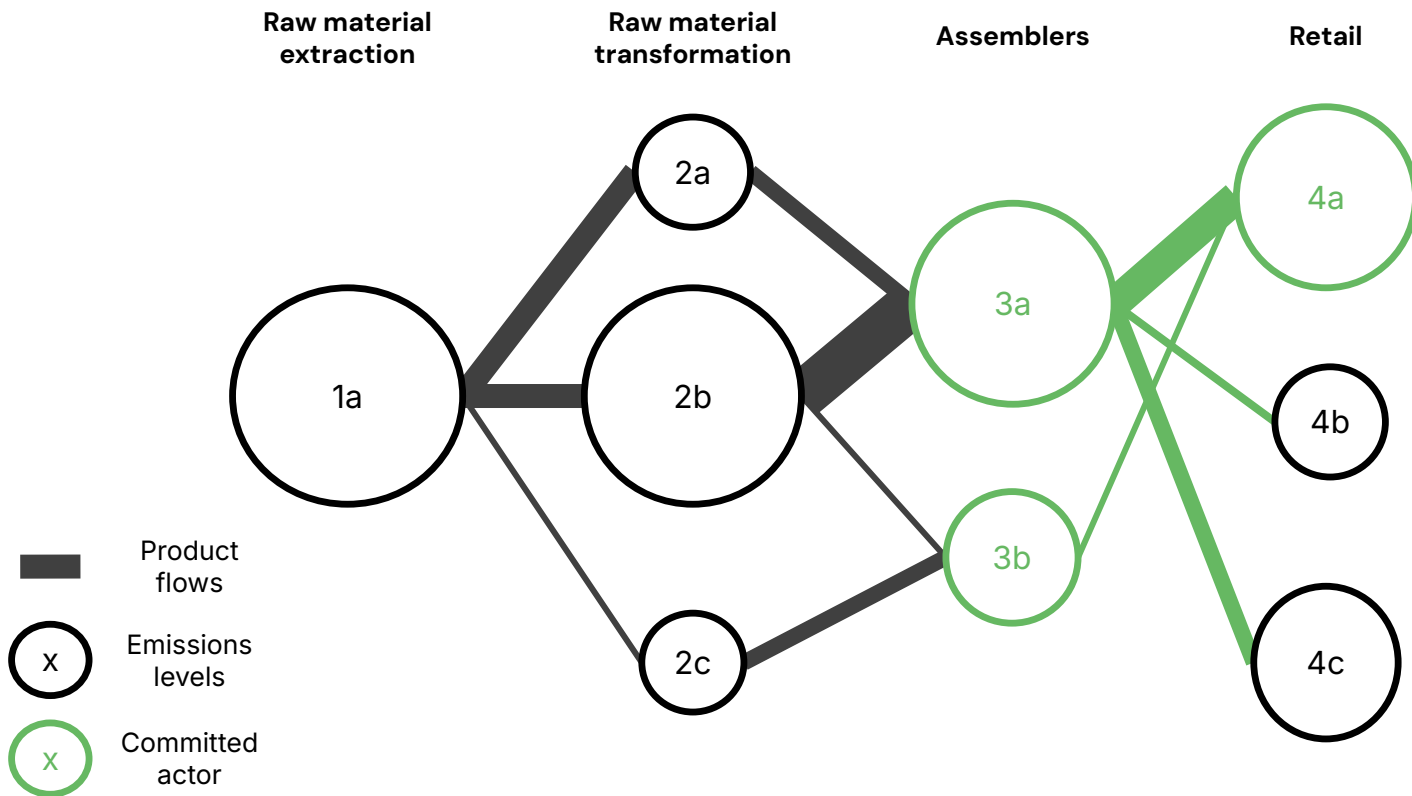
Model overview

Equilibrium disruption : retailer 1a requires company wide environmental commitments



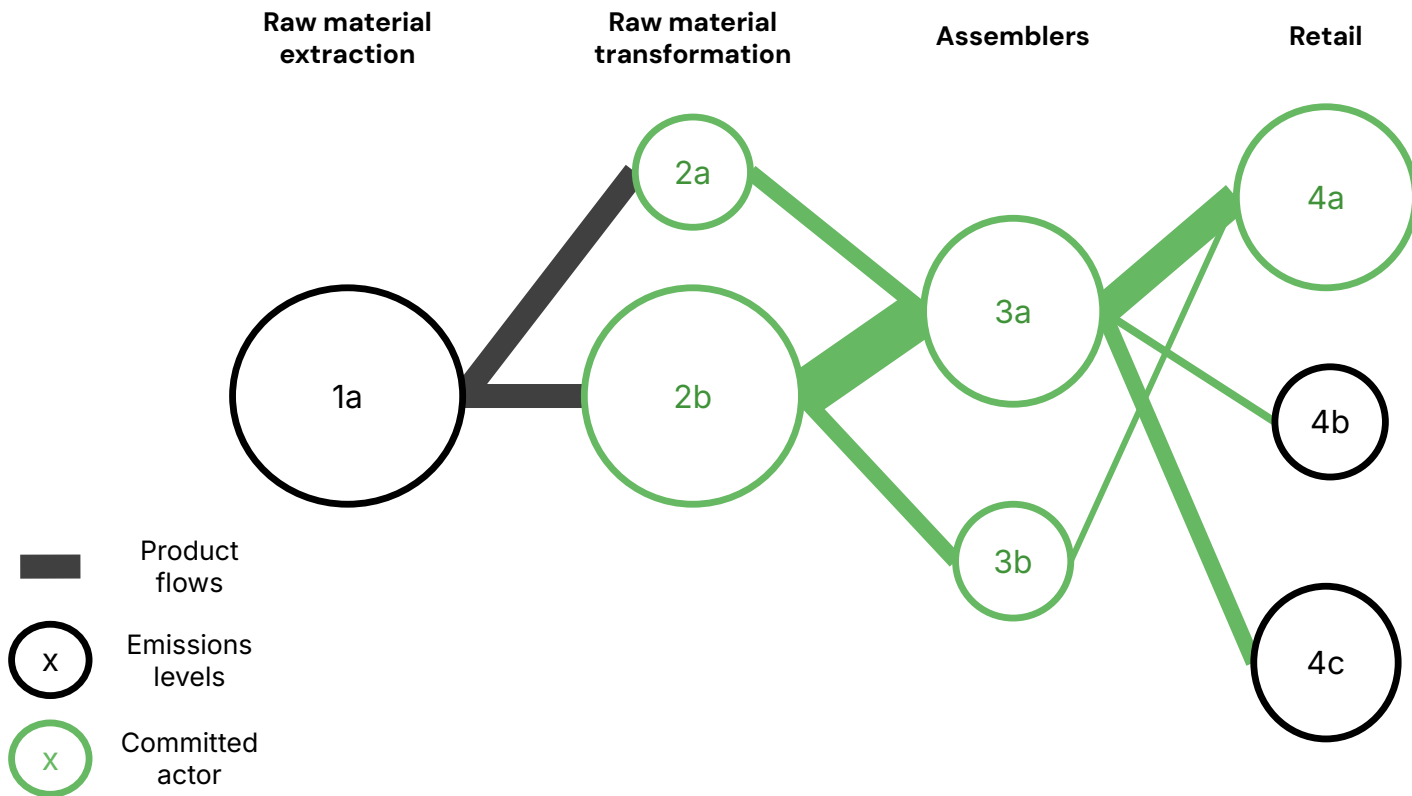
Model overview

Updated equilibrium at $t=1$



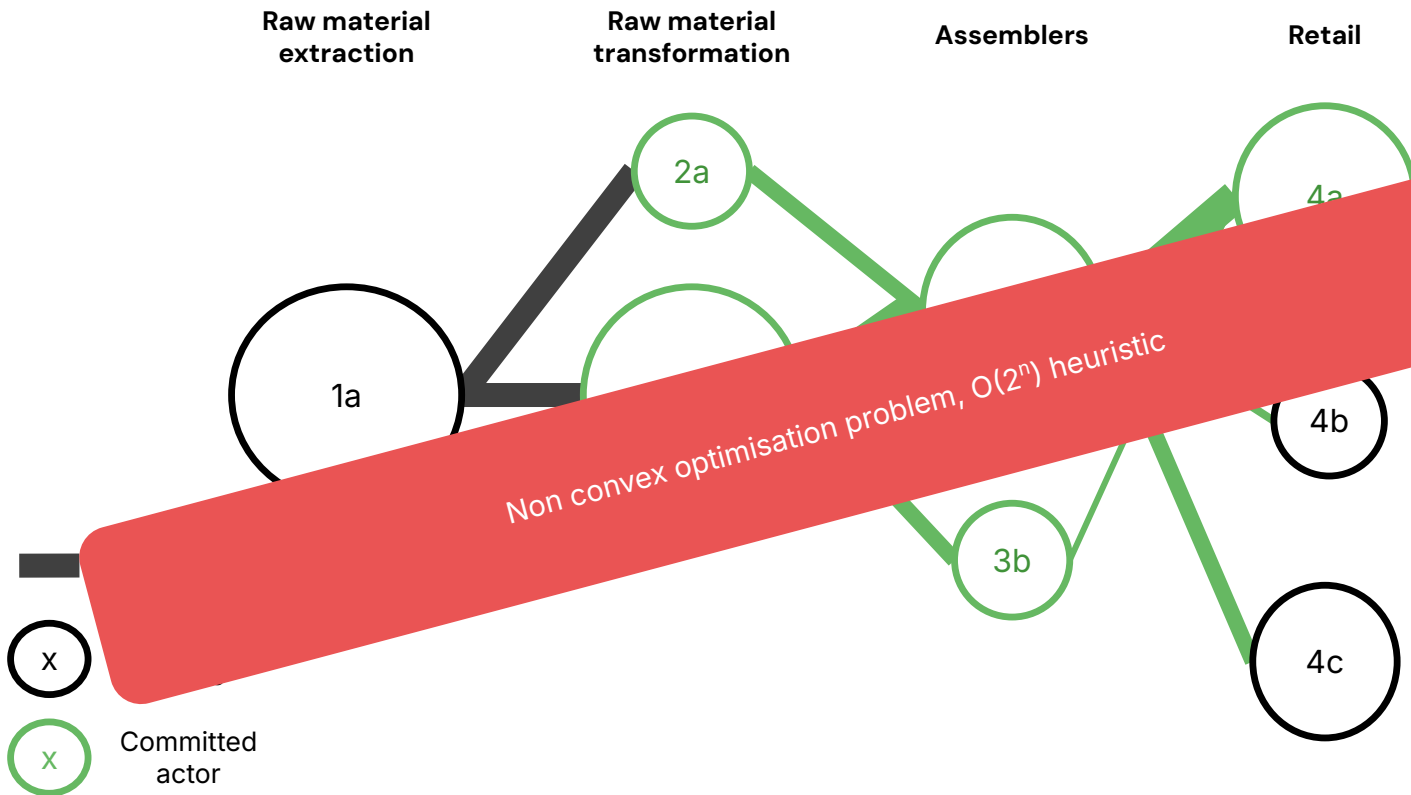
Model overview

Updated equilibrium at $t=2$



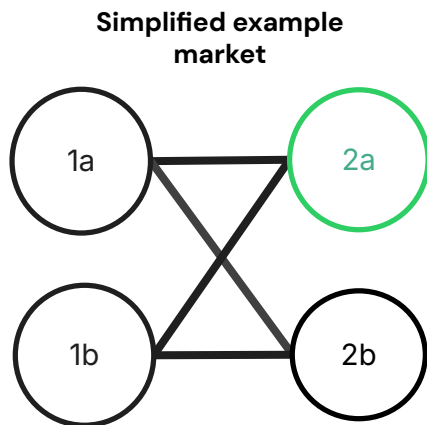
Model overview

Updated equilibrium at $t=2$



Backward induction solving

An alternative, polynomial approach to solving the problem - a 2-2 example



Two step game

1. Choose to stay conventional (c) and renounce selling to 1, or commit to company wide ESG label (s) and pay ESG costs across their production
2. Optimise sold quantities to each buyer given strategy choices

Simplified iterative partial optimum exploration routine

- Consider **(s,s)** initial state
- Order suppliers by **decreasing ESG costs**
- While the sustainable supplier with the highest ESG cost has an incentive to **unilaterally switch strategies**
 - Switch the strategy for the supplier
- The resulting state is a **Nash equilibrium**

Backward induction solving

An alternative, polynomial approach to solving the problem - a 2-2 example

Table 4.1: Optimal profit outcomes (Π_3, Π_4) for each supplier under all strategy combinations.

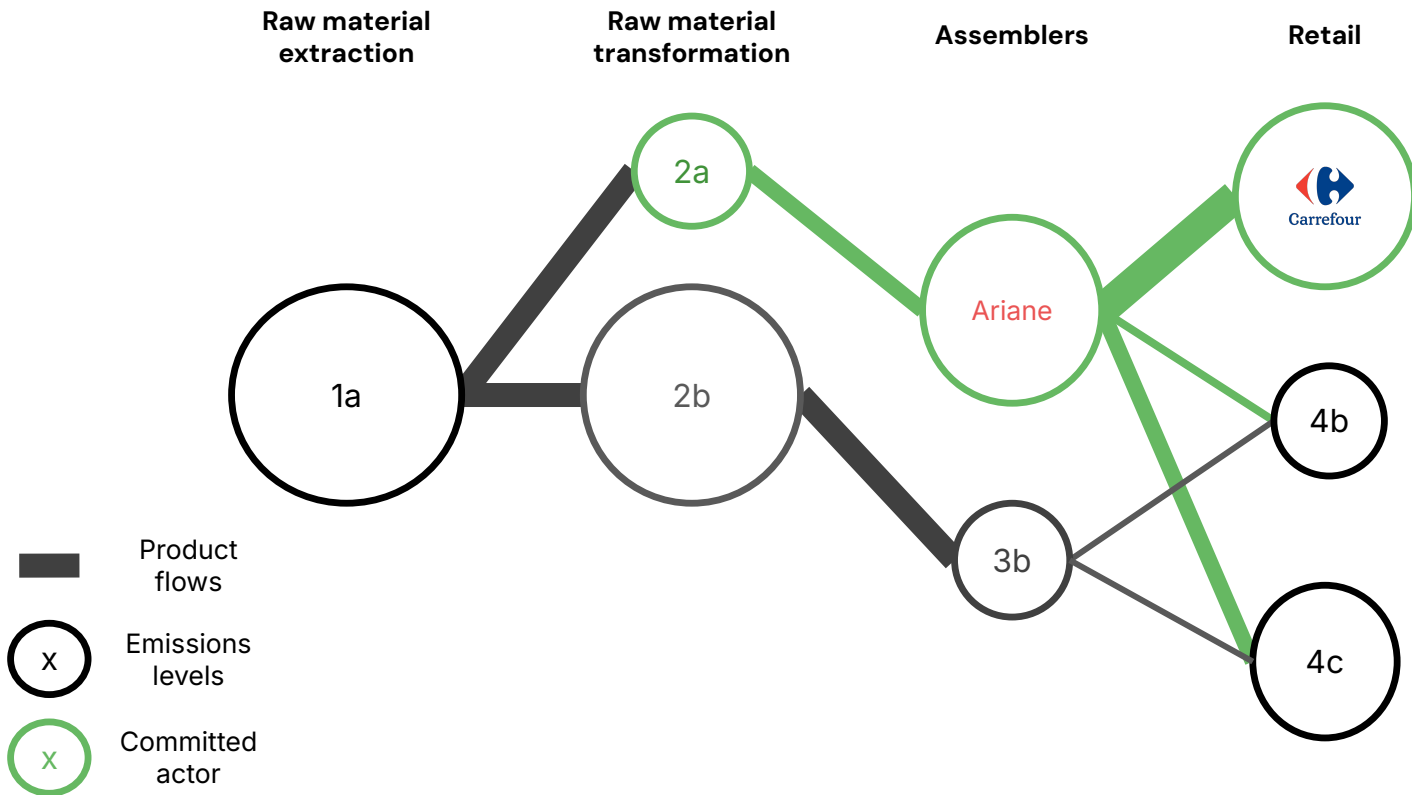
Strategy for supplier 4 → supplier 3 ↓	c	s
c	$\left(\frac{(a_2 - c)^2}{9b_2}, \frac{(a_2 - c)^2}{9b_2} \right)$	$\left(\frac{(a_2 - c + d_4)^2}{9b_2}, \frac{(a_2 - c - 2d_4)^2}{9b_2} + \frac{(a_1 - c - d_4)^2}{4b_1} \right)$
s	$\left(\frac{(a_1 - c - d_3)^2}{4b_1} + \frac{(a_2 - c - 2d_3)^2}{9b_2}, \frac{(a_2 - c + d_3)^2}{9b_2} \right)$	$\left(\frac{(a_1 - c + d_4 - 2d_3)^2}{9b_1} + \frac{(a_2 - c - 2d_3 + d_4)^2}{9b_2}, \frac{(a_1 - c + d_3 - 2d_4)^2}{9b_1} + \frac{(a_2 - c + d_3 - 2d_4)^2}{9b_2} \right)$

Simplified iterative partial optimum exploration routine

- Consider **(s,s) initial state**
- Order suppliers by **decreasing ESG costs**
- While the sustainable supplier with the highest ESG cost has an incentive to **unilaterally switch strategies**
 - Switch the strategy for the supplier
- The resulting state is a **Nash equilibrium**

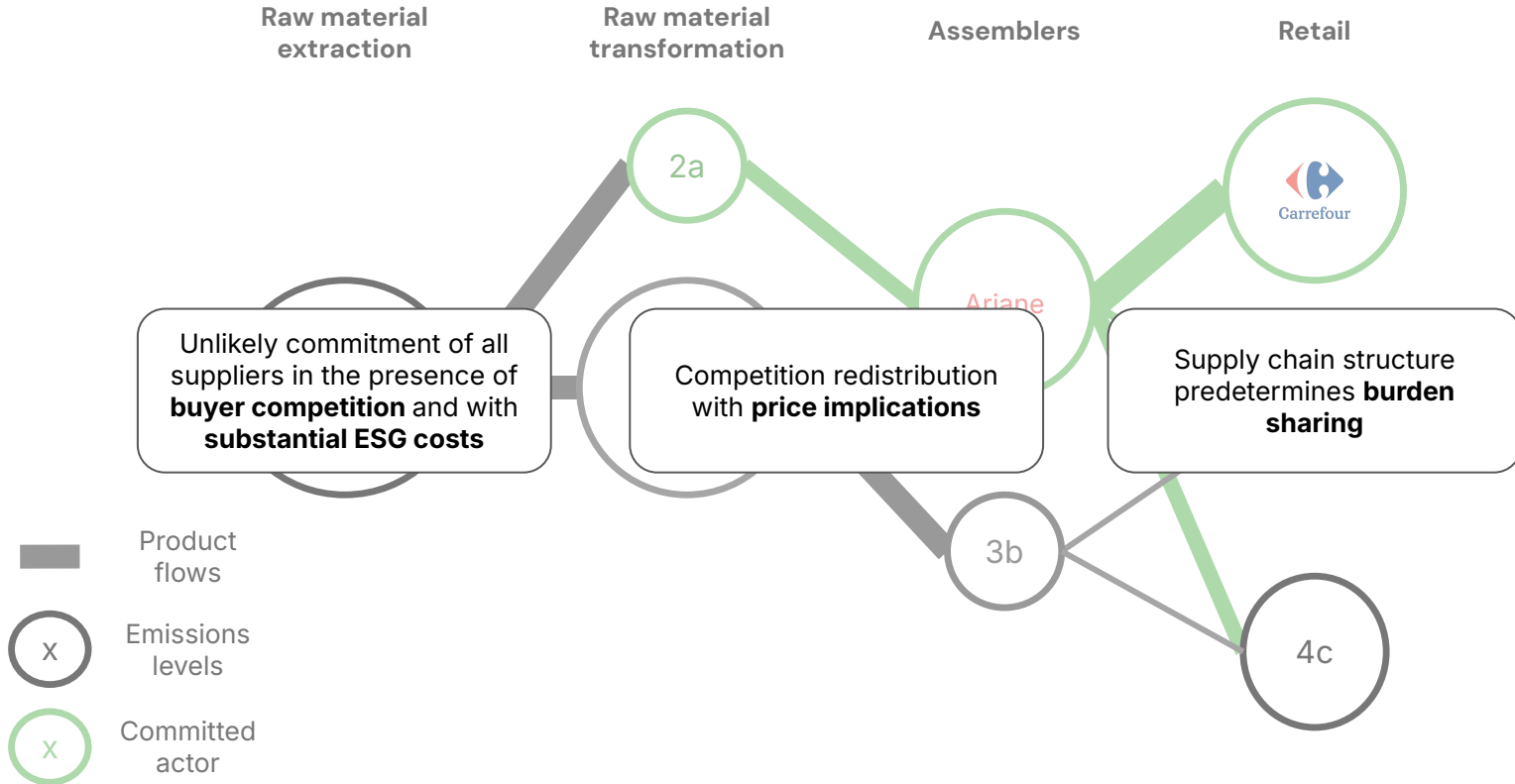
Preliminary model insights

Coercive practices limitations and impacts



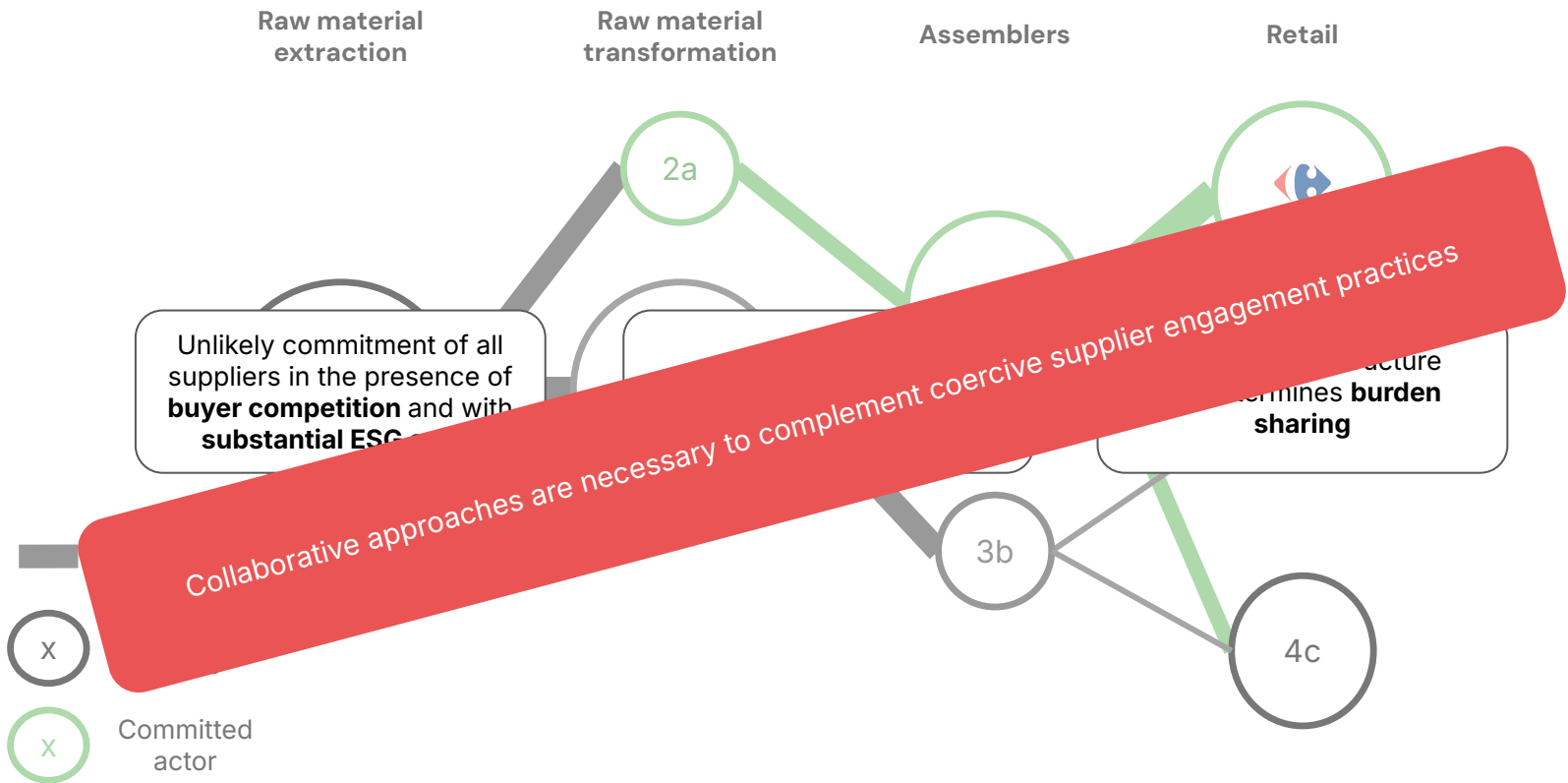
Preliminary model insights

Coercive practices limitations and impacts



Preliminary model insights

Coercive practices limitations and impacts



Chapter 5

Alternative allocation approaches for
Scope 3 upstream emissions: a
principled approach

Marine Kohler, David Lowing



Research questions

Beyond coercive practice, what could be done?



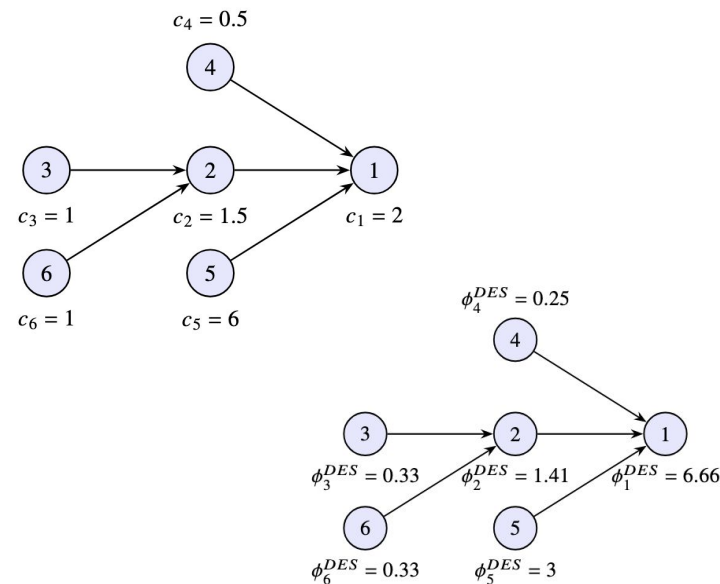
- How could a company **systematically and fairly subsidize** the emissions abatement of their suppliers?
- What would the **resulting burden sharing** be if upstream subsidies were applied at a large scale?

Defining and operationalizing fair allocation

Bridging two complementary literature strands



River problem allocation functions



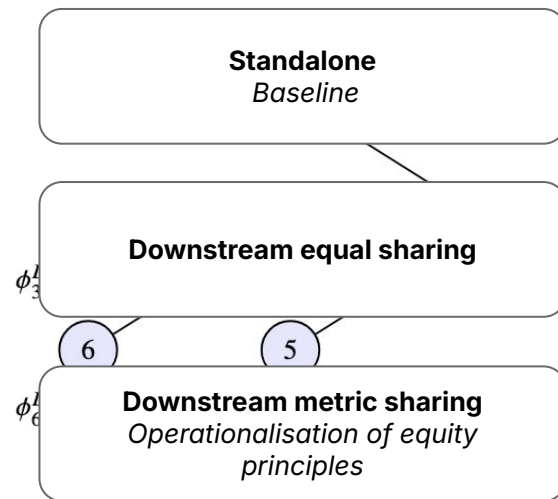
Robiou Du Pont, Y., Dekker, et al. (2025). Nature Communications, 16(1), 8020; Bai, X., Hasan, S., et al. (2024), Nature Sustainability, 7(2), 108–119; Li, W., G. Xu, and R. van den Brink (2023), Social Choice and Welfare, 61, 35–59; Ni, D. and Y. Wang (2007), Games and Economic Behavior, 60, 176–186.

Defining and operationalizing fair allocation

Bridging two complementary literature strands



River problem allocation functions



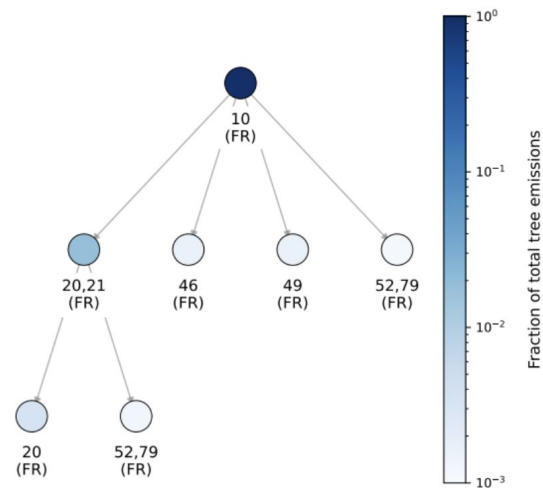
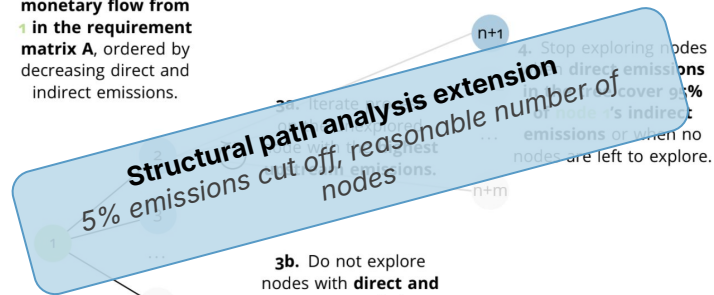
Robiou Du Pont, Y., Dekker, et al. (2025). Nature Communications, 16(1), 8020; Bai, X., Hasan, S., et al. (2024), Nature Sustainability, 7(2), 108–119; Li, W., G. Xu, and R. van den Brink (2023), Social Choice and Welfare, 61, 35–59; Ni, D. and Y. Wang (2007), Games and Economic Behavior, 60, 176–186.

Application

Two challenges for large scale implementation

Estimating plausible supply chain structures

1. Add a node for each country-sector pairs for which there is **non nul monetary flow from** **1 in the requirement matrix A**, ordered by decreasing direct and indirect emissions.



Owen, A., R. Wood, J. Barrett, and A. Evans (2016), Economic Systems Research, 28, 243–272; Lopez-Garcia, P., F. di Mauro, and CompNet Task Force (2015): ECB Working Paper 1764, European Central Bank.

Application

Two challenges for large scale implementation

Mapping equity-enabling metrics

CompNet The Competitiveness Research Network

Responsibility

Direct emissions multiplied by firm age

Economic contribution

Labor productivity, profit margin, operating revenue

Social contribution

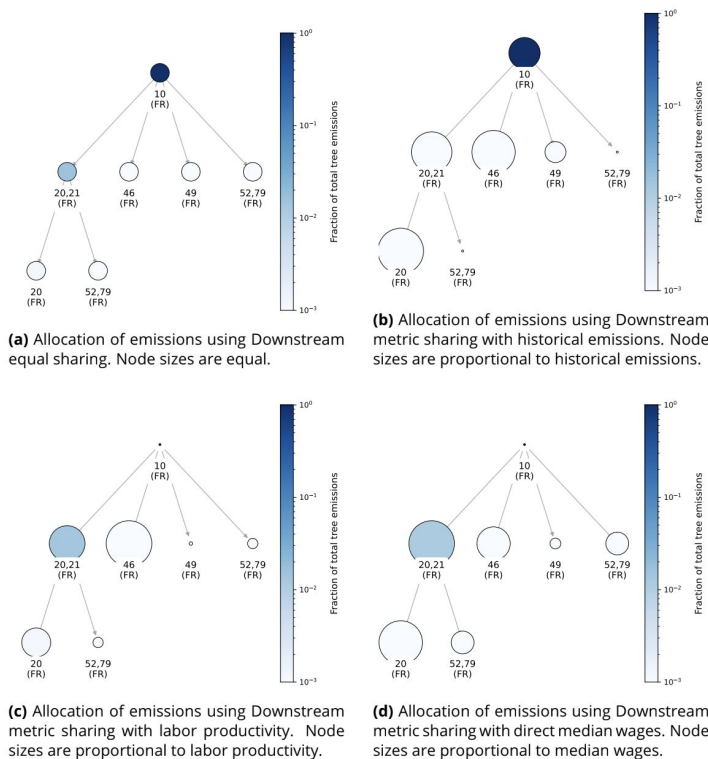
Number of employees, Labor cost, Median wage

Green incentive

Direct emissions, direct emissions intensity

Capability

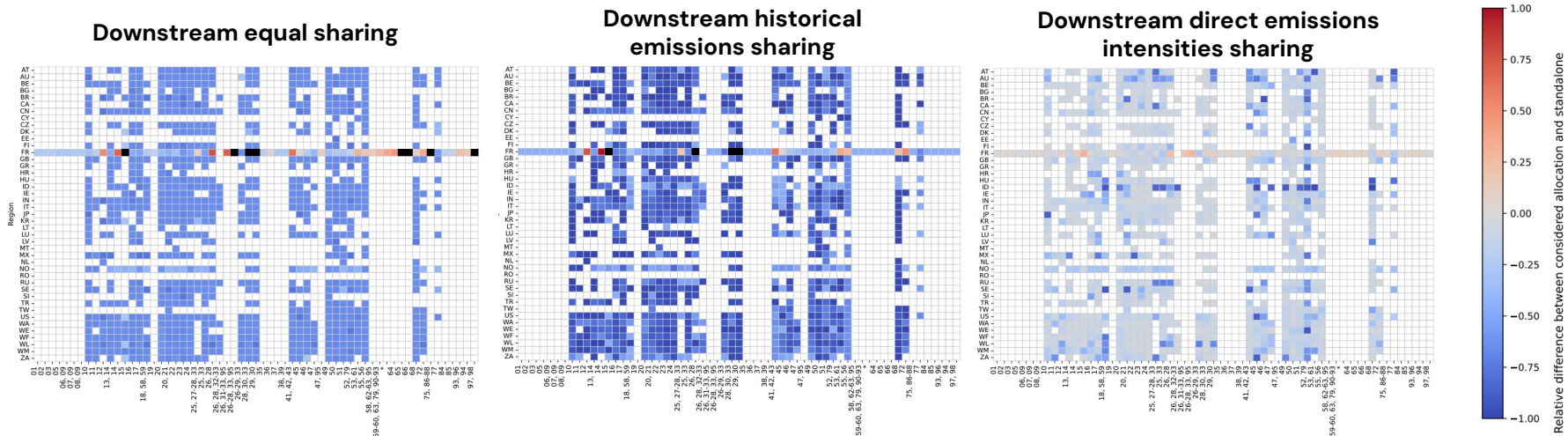
Liquidity ratio, debt ratio



Owen, A., R. Wood, J. Barrett, and A. Evans (2016), *Economic Systems Research*, 28, 243–272; Lopez-Garcia, P., F. di Mauro, and CompNet Task Force (2015): ECB Working Paper 1764, European Central Bank.

What if all French companies followed our allocation rules?

A diversity of outcomes for downstream equal and metric sharing



The downstream logic dominates compared to equity principles

Uneven domestic consequences

Magnitude of removed upstream responsibility significantly depends on equity metric

Conclusions and perspectives

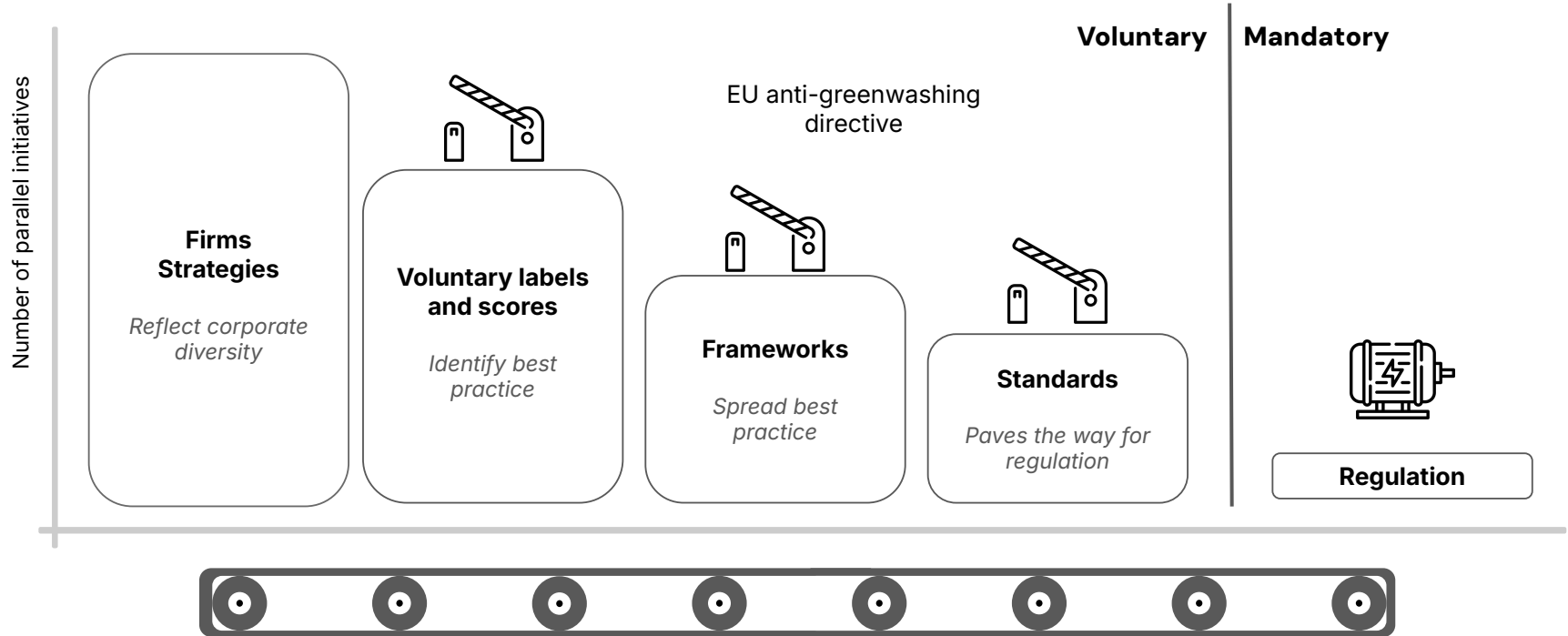
Overview of contributions

Output with a combined academic and industrial interest

	Academic	Industrial
Chapter 2	Critical climate performance indicator market review	Competitor benchmark, guidelines for label design
Chapter 3	Firm-level EKC to firms	Emissions benchmarks
Chapter 4	ESG labels and heterogenous costs Cournot model	Influence proxy estimation algorithm
Chapter 5	Large-scale allocation of scope 3 emissions	Extended structural path analysis

Perspectives

Do corporate climate ratings advance or hinder the Net Zero transition?



Adapted from Hale, T. (2022, December). The net-zero governance conveyor belt. Kleinman Center for Energy Policy. Evidence for stakeholder expectation management can be found on the CDP, GHG Protocol and SBTi websites.

Future research

Some extensions and a range of possible directions

